Project Report: Comparative Analysis of Abstractive Summarization Models

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Selected Task: Text Summarization
Colab Link:
https://colab.research.google.com/drive/1xiMyfaQ-RYvEkLcp4g3o02ckj-S0JfD1?usp=sharin
g
Github Link: https://github.com/Rishith-S/dl-hw2

1. Introduction

Topic Selected: Summarization
Dataset Selected:I utilized the SAMSum Dataset, a corpus of messenger-like conversations with human-written summaries. This dataset presents a unique challenge as models must interpret informal language, slang, and speaker turns.
Dataset link : https://huggingface.co/datasets/knkarthick/samsum

2. Approach & Model Architectures

To satisfy the requirement of comparing multiple architectures, I selected three distinct Encoder-Decoder Transformer models. I evaluated them in two states: **Zero-Shot (Base)** and **Fine-Tuned**.

1. **T5-Small (Baseline):** A "Text-to-Text Transfer Transformer" that treats summarization as a translation task. It is lightlight (60M parameters).
2. **BART-Base (Standard):** A Denoising Autoencoder with a bidirectional encoder (like BERT) and an autoregressive decoder (like GPT). It is highly effective for text generation.
3. **DistilBART-CNN-6-6 (Optimized):** A distilled version of BART-Large. It mimics the performance of larger models while being significantly smaller (230M parameters) and faster.

Training Configuration:

I fine-tuned all three models using the Hugging Face Seq2SeqTrainer.

- **Training Data:** 2,000 samples from the SAMSum train split.
- **Hyperparameters:** Learning Rate: 2e-5, Batch Size: 4, Epochs: 1.
- **Hardware:** Training was performed on a T4 GPU.

3. Evaluation & Results

I evaluated performance using **ROUGE** (Recall-Oriented Understudy for Gisting Evaluation), specifically **ROUGE-1** (unigram overlap), which is the standard metric for summarization content coverage.

A. Quantitative Comparison (Base vs. Fine-Tuned)

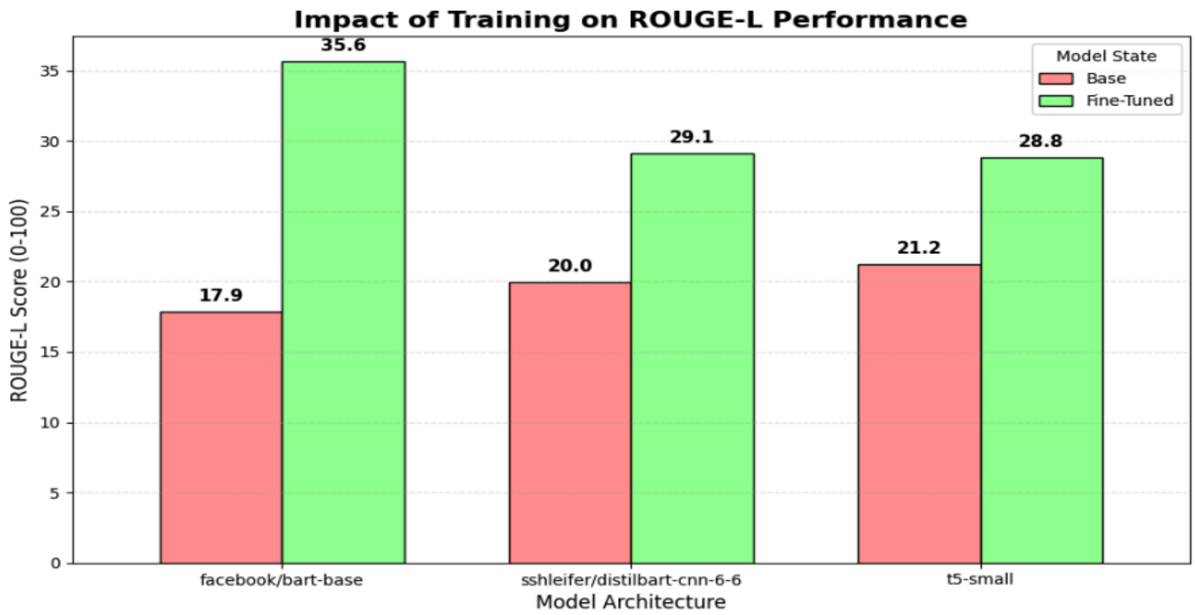
The table below demonstrates the impact of training on domain-specific data.

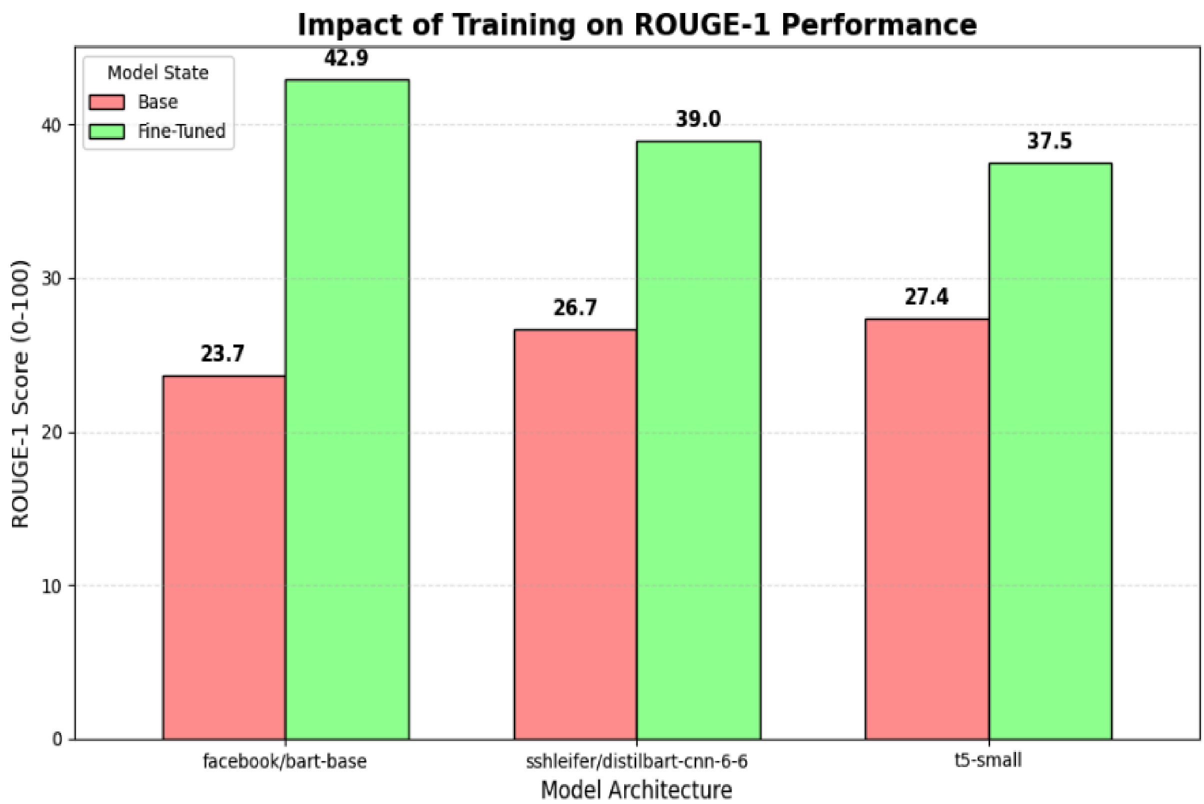
Model Architecture	Metric	Base Score (Zero-Shot)	Fine-Tuned Score	Absolute Improvement
facebook/bart-base	ROUGE-1	23.7	42.9	+19.2
	ROUGE-2	7.8	19.1	+11.3
	ROUGE-L	17.9	35.6	+17.7
sshleifer/distilbart-cnn-6-6	ROUGE-1	26.7	39.0	+12.3
	ROUGE-2	8.4	18.1	+9.7
	ROUGE-L	20.0	29.1	+9.1
t5-small	ROUGE-1	27.4	37.5	+10.1
	ROUGE-2	7.4	15.2	+7.8
	ROUGE-L	21.2	28.8	+7.6

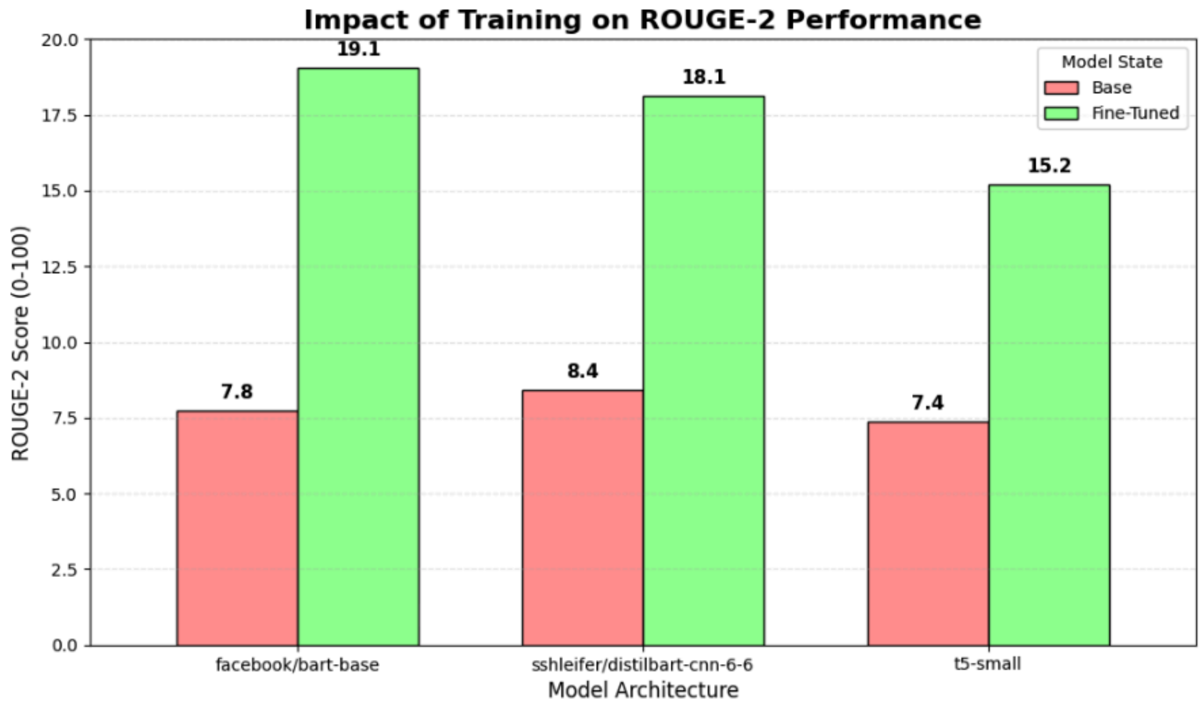
Observation: The results demonstrate a consistent and significant improvement across all three architectures after fine-tuning on 2,000 samples of the SAMSum dataset. The facebook/bart-base model emerged as the top performer, showing the highest responsiveness to domain adaptation.

B. Performance Visualizations

Figure 1: Comparison of Fine-Tuned Architectures







4. Qualitative Analysis

I tested the models on a custom sample to verify readability.

Input Dialogue:

> Doctor: Hi, how are you feeling today?
>
> Patient: I have a terrible headache and a sore throat.
>
> Doctor: How long have you had these symptoms?
>
> Patient: About three days now.
>
> Doctor: Okay, let me check your temperature. It seems you have a slight fever.
>
> Patient: Do I need antibiotics?
>
> Doctor: Not yet. I'll prescribe some rest and fluids first. Come back if it gets worse.

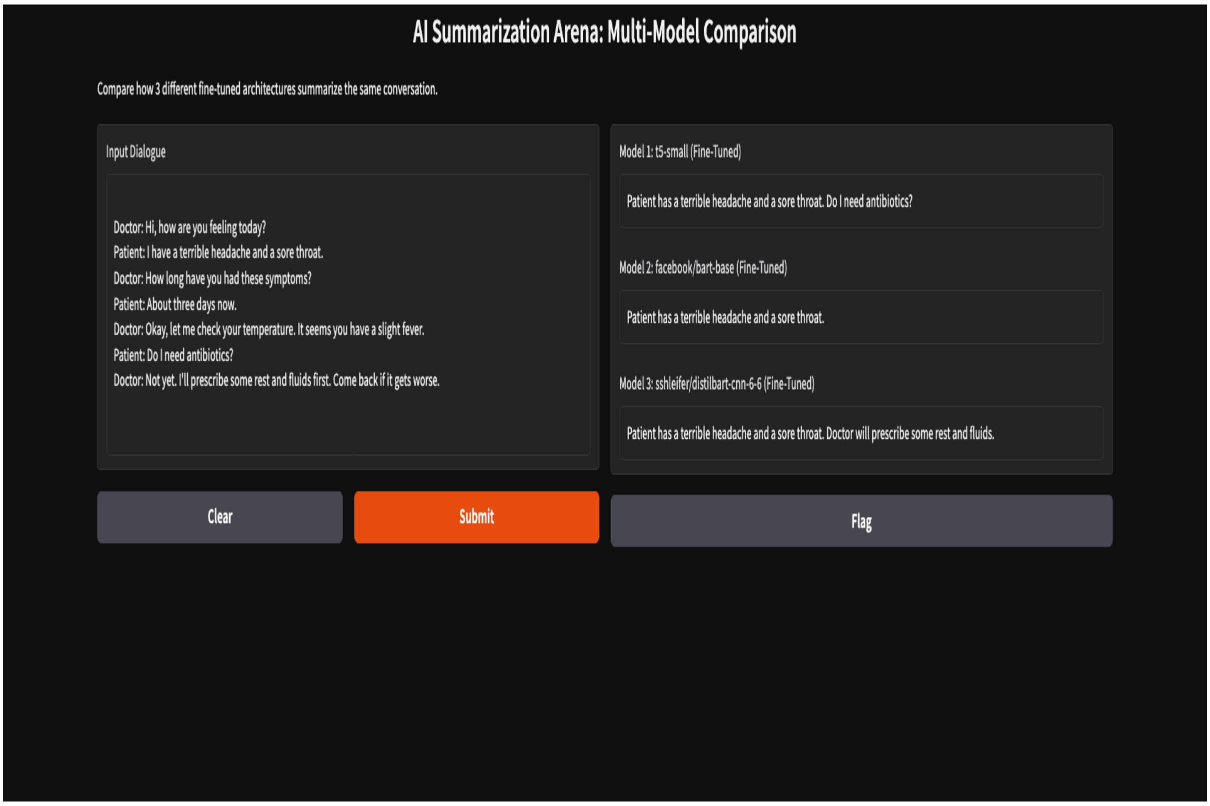
Model Output:

“Patient has a terrible headache and a sore throat. Doctor will prescribe some rest and fluids.”

Verdict: The model correctly identified the speakers, Patient and Doctor, and the prescription.

5. Application UI

To make the system accessible, I developed an interface using **Gradio**. The UI loads all three fine-tuned models and allows users to compare their summaries side-by-side in real-time.Interface Screenshot:



6. Conclusion

This project successfully demonstrated that fine-tuning smaller, efficient Transformer models significantly outperformed the zero-shot baselines.